

SOC4190 BLOCKCHAIN BASICS & PLATFORMS

Mining: Who Writes the Block

Mode of teaching: **Online, asynchronous**

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Where We Stopped

My video

From the last lectures you need

block header and its six fields · **block hash** (SHA-256 twice; blockhash.py) · the five promises, above all
one-way · **Merkle root** · **miner** · **transaction fee** · **mempool** · **double spending** · **little-endian**

Field	Bytes	In plain words	
Version	4	which set of rules the block follows	✅
Previous block hash	32	the fingerprint of the block before — the link of the chain	✅
Merkle root	32	one fingerprint for all the transactions in the body	✅
Timestamp	4	when the block was made	✅
Bits (target)	4	how hard it was to make this block	❓
Nonce	4	a counter used while making the block	❓

What is the work of making a block?

The Block With Our Payment

Stop	Today's question	What we know	
The writer	Who gets to write the next block?	—	?
The nonce	What is the nonce, exactly?	—	?
The bits	What do the bits say?	—	?
The pay	What does the miner get — and where do new coins come from?	—	?
The rhythm	Why about one block every ten minutes?	—	?
The danger	What if two miners win at once — or one gets too strong?	—	?

My video

Who wrote this block?

BLOCK 968,391 · 24 Sep 2026, 11:49 UTC

HEADER · 80 BYTES

version	0x22fce000
previous block hash	…ab35007dbe
merkle root	…26ba124a93
time	2026-09-24 11:49:38
bits	0x17021ec5
nonce	3,682,494,035

its hash: 0000000000000000000006a6e…8da3

BODY · 4,117 TRANSACTIONS

- **#0 · the miner's own transaction**
  pays the miner
- **#1 … #450**
  other payments
- **#451 · our payment d9f91d…6dd6**
  fee 1,356 sat (Lecture 6)
- **#452 … #4,116**
  other payments

the body is summed up in the header by the Merkle root

A Story: Who Writes the Next Page?

Anyone may write the next page, but every try must cost real work.

My video

300 students, one shared notebook, thousands of copies: who may write the next page?

THE TEACHER DECIDES

one person keeps the notebook:
an intermediary again

TAKE TURNS

who is next? and who is
a real student?

A VOTE

one person online can pretend
to be a thousand people

Lecture 1: everyone keeps a copy of the same notebook. But somebody must write the **next page** — choose which payments go in, and in which order. If the wrong person writes it, they can write their own payment twice.

Bitcoin's answer

Let anyone try — but make every try cost real work, and pay the winner. Pretending to be a thousand people does not help: a thousand fake names do no more work than one.

Role of a Miner

My video

A miner ⛏️ is not just a participant; they are a vital organ of the decentralized system. They simultaneously serve as the network's validating accountants 📃✅, its security guards 🛡️, its currency issuers 💰, and, in many ways, its decentralized governance body 🏛️. Their role is not passive; it is an active, economically-driven function ⚡📈 that provides the trust 🤝 and security 🔒 a decentralized ledger needs to operate.

The role of a miner is far more complex than simply "running a computer" 💻. They are the accountants who validate transactions 📃, the security guards who protect the ledger's history 🛡️📖, the mint that issues new currency 🪙, and the governors who help shape the network's future 🗺️. It is a system of elegantly designed incentives 💡 where the miner's pursuit of profit 🤑 is precisely what provides the security 🔒, integrity ✅, and operational power ⚙️ of the entire decentralized network 🌐.

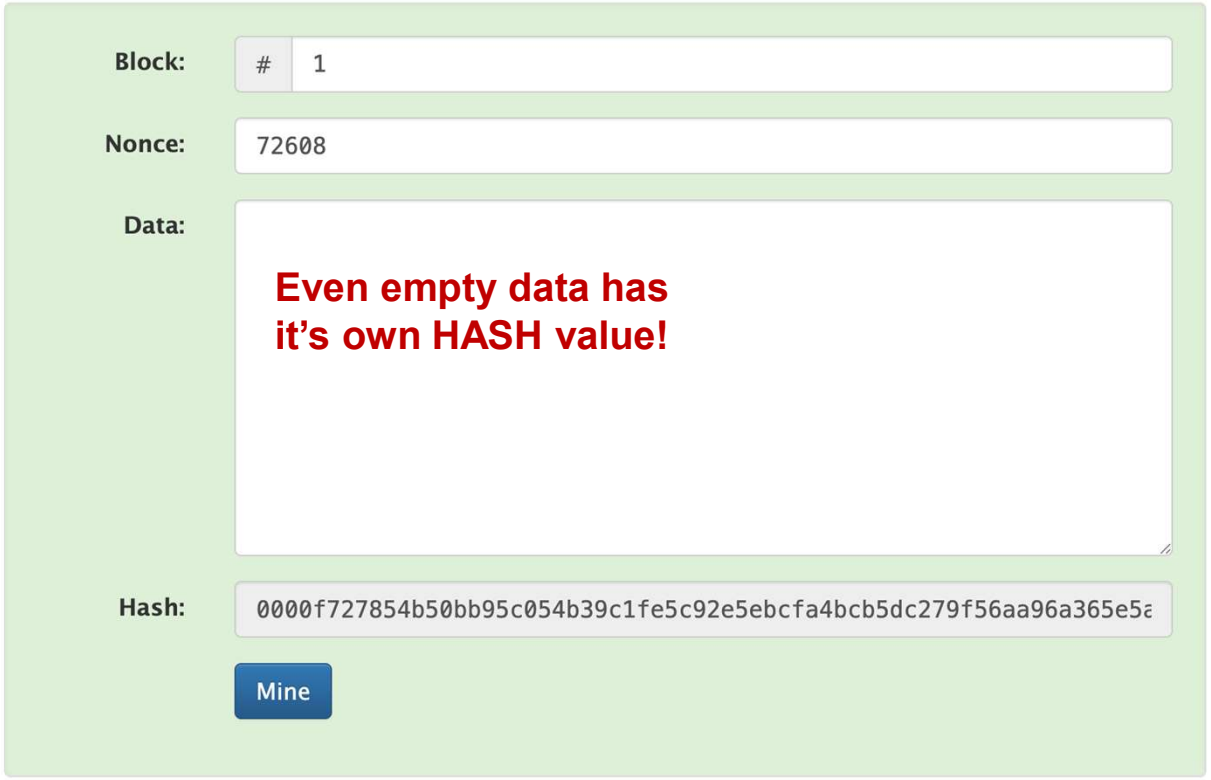


Miners are essential for:

- **Decentralization**: No single entity controls the ledger.
- **Security**: Making it computationally expensive to alter past transactions.
- **Consensus**: Ensuring all nodes agree on the state of the blockchain.

LIVE DEMO: The Puzzle — Make the Hash Start With Zeros

https://andersbrownworth.com/blockchain/block



My video

Finding a small hash takes millions of tries; there is no shortcut.

New Words: Mining and Proof of Work

My video

New word · Mining

Building a block and trying nonce after nonce until the block's hash is small enough. The first miner to succeed adds the block to the chain and is paid.

every miner in the world runs this loop at the same time, on its own block



Mining is the loop; proof of work is the small hash it leaves behind.

New word · Proof of work

A block hash small enough to show that a great many tries were made. Nobody can fake it, and anyone can check it with one single hash.



Hard to Find, Easy to Check

https://andersbrownworth.com/blockchain

My video

Why this protects the past

Lecture 3: change one old payment and that block's hash changes — and every later block's link breaks. Now add the work: the cheater must redo the puzzle for that block and every block after it, faster than everyone else.

	The miner	Every other node
Job	find a nonce that makes the hash small enough	check that it does
Work	about 5.7×10^{23} hashes for block 968,391	**one** hash
If someone cheats	a block with a big hash is thrown away by everyone	the cheat is caught at once

The number of tries

How 5.7×10^{23} is computed: the next part of today.

Writing a block is hard; checking it is easy: that asymmetry is the security.

New Word: Target

New word · Target

A 256-bit number fixed by the rules of the moment. A block is valid only if its hash, read as a number, is **below** the target.

Read left to right, like two prices: the first 19 digits are equal; then the hash has **0** where the target has **2**. The hash is smaller ✓

My video

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HEADER · 80 BYTES

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- **#452 … #4,116** — other payments

the body is summed up in the header by the Merkle root

A block counts only if its hash is below the target.

```
hash   0000000000000000000006a6ecfff40…
target 0000000000000000000021ec5000000…
```

Bits: The Target in Four Bytes

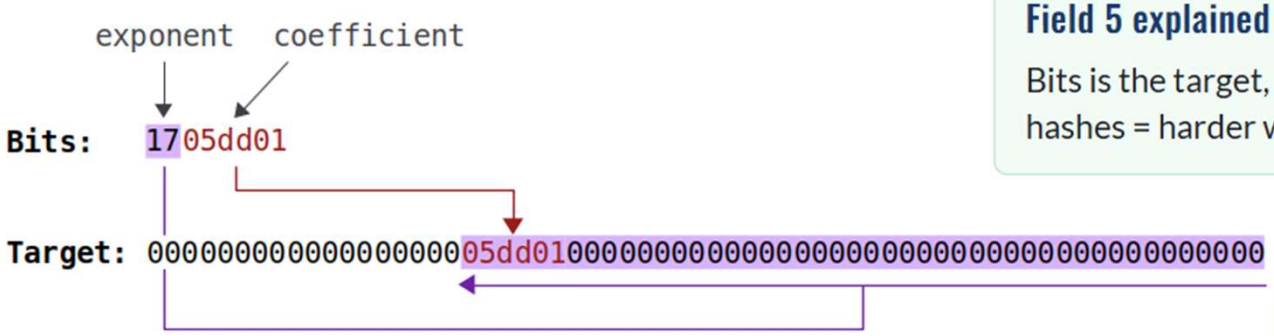
CALCULATE The target has 64 hexadecimal digits; the header has room for 4 bytes. So it is written short.

```
in the header    c5 1e 02 17   (little-endian)
backwards        17 02 1e c5
first byte       0x17 = 23: the length in bytes
other three      021ec5: the digits

target = 021ec5 followed by 40 zeros
       = 0000000000000000000021ec5…
```

My video

The first byte says how long the number is: 23 bytes. The other three are its first digits. Fill the rest with zeros — 40 hexadecimal zeros — and write it as 64 digits.



Field 5 explained

Bits is the target, packed into 4 bytes. Lower target = fewer good hashes = harder work.

Bits = the target, written as a length and three digits.

LIVE DEMO: The Nonce, Checked

```
$ python3 blockhash.py header_968391.hex
80 bytes  block hash: 0000000000000000000006a6ecfff40e7f73228dd2302732b917e38769c398da3
$ python3 blockhash.py header_968391_nonce_plus1.hex
80 bytes  block hash: 483cd7a0201abd010d9790198c13df3c4094559c48fd301bca9ef03da8e93437
```

LIVE DEMO Lecture 4's `blockhash.py`, on block 968,391's header

My video

HEADER · 80 BYTES

version	0x22fce000
previous block hash	…ab35007dbe
merkle root	…26ba124a93
time	2026-09-24 11:49:38
bits	0x17021ec5
nonce	**3,682,494,035**

its hash: 0000000000000000006a6e…8da3

With nonce **3,682,494,035** the hash starts with 20 zeros: below the target ✓.

One more, and not a single zero is left ✗.

BODY · 4,117 TRANSACTIONS

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the body is summed up in the header by the Merkle root

The nonce is the winning ticket: one number that makes the hash small enough.

BLOCK 968,391 · 24 Sep 2026, 11:49 UTC

Four Billion Tries Are Not Enough

CALCULATE The nonce has 4 bytes. Is that enough room to find a block?

$$\frac{5.7 \times 10^{23}}{4.29 \times 10^{9}} \approx 1.3 \times 10^{14}$$

- **Total guesses needed:** 5.7×10^{23} (from above)
- **Guesses per nonce run:** 4.29×10^{9} (the 2^{32} possible nonces)

```
possible nonces     2^32 = 4,294,967,296
tries needed        about 5.7 × 10^23
                    (block 968,391)
the nonce runs out  about 1.3 × 10^14 times
```

My video

So miners change other bytes too

When all 4 billion nonces fail, the miner changes a few free bytes inside its own transaction #0. That changes the Merkle root (Lecture 4), so the header is new — and 4 billion fresh nonces are ready. The time field moves forward as well.



Four billion nonces run out fast; miners also reshape transaction #0 and start again.

New Word: Difficulty



Block 968,391 needed about **1.3×10^{14}** times more tries than the first blocks of 2009 — about 5.7×10^{23} in all. Today's difficulty, after the last adjustment: **140,577,680,051,920**.

My video

📘 New word · Difficulty

How many times harder today's target is than the first one of 2009: the first target divided by today's. Every block's header carries it in its bits.

```
first target  00000000ffff00000000…   (bits 0x1d00ffff)
block 968,391 000000000000000000021e…   (bits 0x17021ec5)

difficulty = first ÷ today's = 132,757,073,449,488
```

Difficulty: how many times harder than the first block.

New Word: Hashrate



My video

New word · Hashrate

How many hashes a machine, or the whole network, computes per second.

```
this laptop     1,530,978 block hashes per second
the network     1.0 × 10^21 per second

block 968,391 alone on the laptop:
  5.7 × 10^23 ÷ 1.5 × 10^6 ≈ 11.8 billion years
the whole network:  ≈ 9 minutes
```

A laptop alone would wait billions of years; the whole network needs about ten minutes.

The universe is about 13.8 billion years old (a rounded public estimate).
Mining is done by halls full of special machines that do nothing else.



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Three Questions Gone

SpiderPool is the name of a large **Bitcoin mining pool**—a group of miners who combine their computing power to increase their chances of successfully mining a block.

The writer is whoever finds the nonce first; bits and nonce are explained. Now: SpiderPool spent this work on block 968,391 — what did it get?

What does the winner get?

My video

Stop	Today's question	What we know	
The writer	Who gets to write the next block?	a miner: the first to find a nonce that makes the block hash small enough — proof of work	✅
The nonce	What is the nonce, exactly?	the 4-byte counter the miner changes until the hash is below the target	✅
The bits	What do the bits say?	the target, written in 4 bytes; the difficulty says how much smaller it is than the first one	✅
The pay	What does the miner get — and where do new coins come from?	—	❓
The rhythm	Why about one block every ten minutes?	—	❓
The danger	What if two miners win at once — or one gets too strong?	—	❓

Three questions gone. Next: the miner's pay.

New Word: Coinbase Transaction

My video

New word · Coinbase transaction

The first transaction of every block, written by the miner. Its input points to no earlier output, so its coins are new. It pays the miner.





every input spends an earlier output (Lecture 6) — except the input of transaction #0

In block 968,391: one output of **318,883,485 sat** to SpiderPool's address, plus 5 notes worth nothing. Not related to the company called Coinbase.

coinbase

Every block starts with the miner paying itself: the coinbase transaction.

New Words: Block Subsidy and Block Reward

My video

📘 New word · Block subsidy

The new coins a block may create, fixed by the rules: 3.125 BTC today.

📘 New word · Block reward

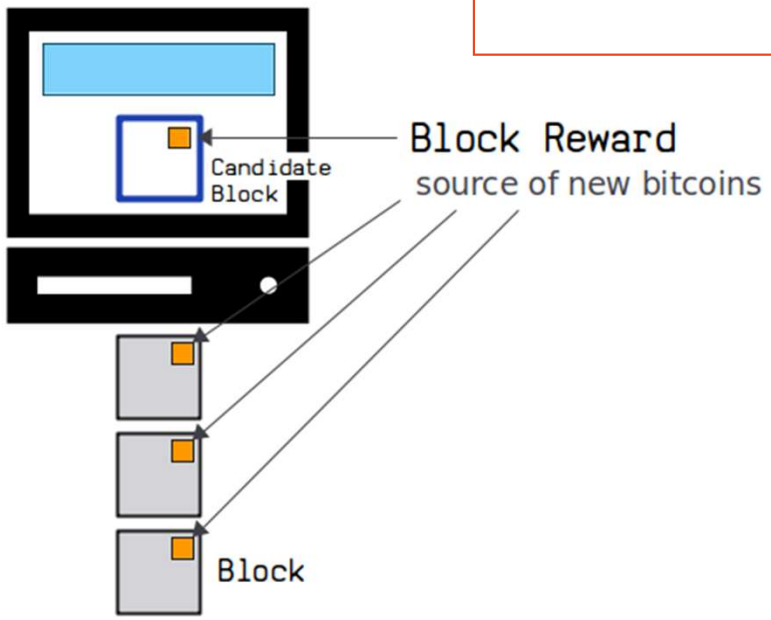
Subsidy + every fee in the block: what the coinbase may pay out, and not one satoshi more.

CALCULATE Block 968,391

```
subsidy              312,500,000 sat
+ fees of 4,116 payments  6,383,485 sat
= block reward       318,883,485 sat ✓
  = 3.18883485 BTC
```

Our 1,356 sat are in there

Lecture 6's fee is one of those 4,116 fees. Every node checks this sum; a coinbase paying more is rejected.



Block reward = new coins + all fees, and every node checks the sum.

CALCULATE: Read the Coinbase

CALCULATE The key of transaction #0 has no signature to carry. What does the miner put there?

```
03 c7 c6 0e 043a0eb56a537069646572506f…
```

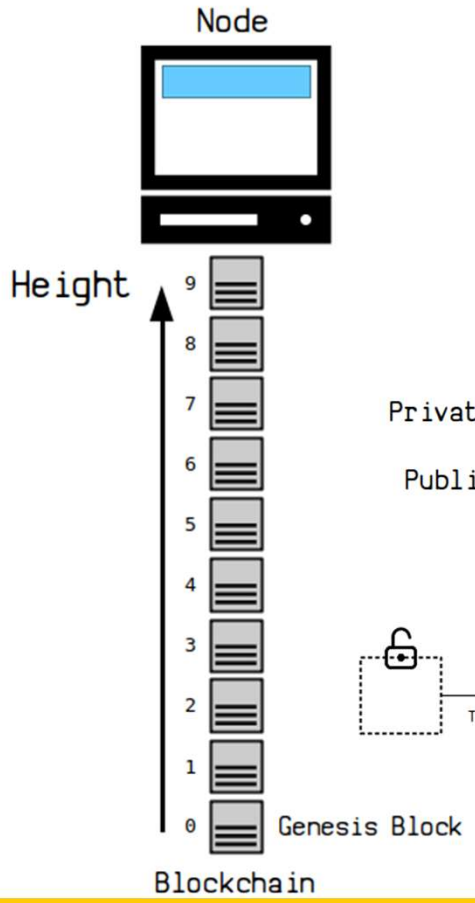
`03`	"3 bytes follow"
`c7 c6 0e`	backwards: `0ec6c7` = **968,391** — the height ✓
readable text	`SpiderPool/1376/`

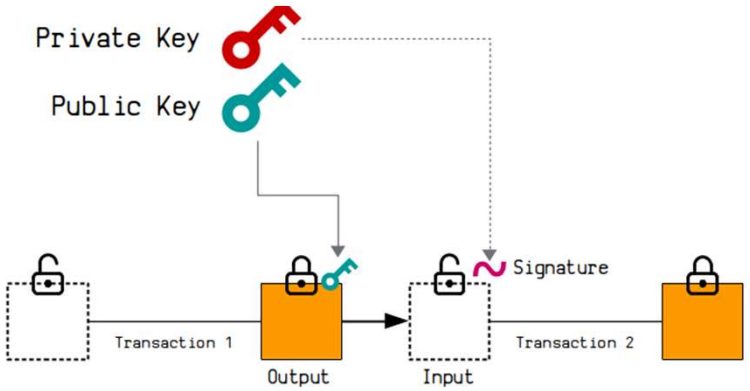
Block 0 did the same

Lecture 3: Satoshi put a newspaper headline into the very first block. It sat exactly here, in the key of transaction #0. Miners still sign their blocks this way.

Transaction #0 carries the height and the miner's own words.

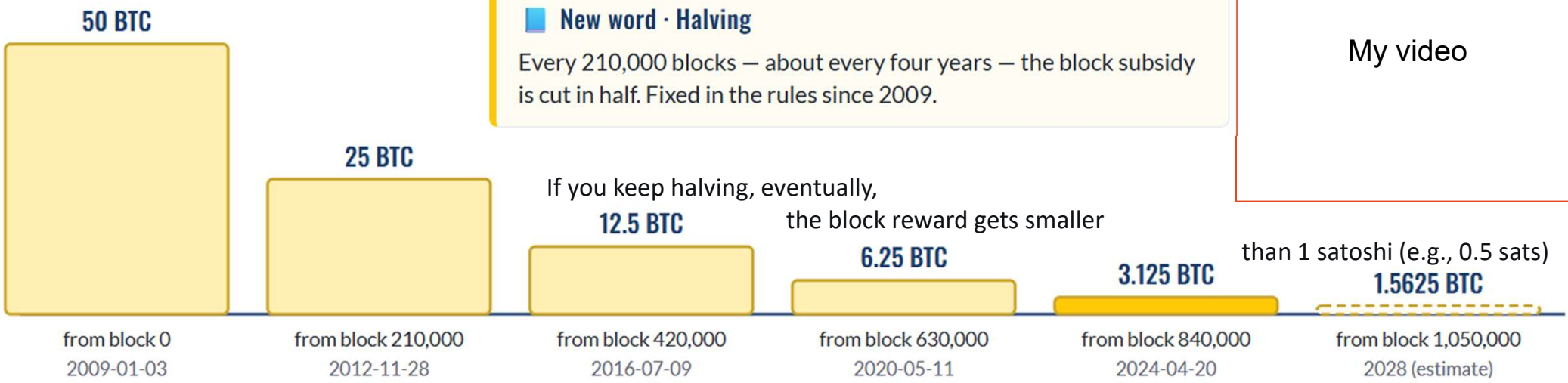
My video





New Word: Halving

My video



New word · Halving

Every 210,000 blocks — about every four years — the block subsidy is cut in half. Fixed in the rules since 2009.

If you keep halving, eventually,
the block reward gets smaller
than 1 satoshi (e.g., 0.5 sats)

Measured: block 839,999 created 6.25 new BTC; block 840,000, four minutes later, only 3.125. The next halving is at block 1,050,000 — 78,129 blocks from now.

Every 210,000 blocks, the new coins per block halve.



HODL is a popular cryptocurrency slang term and long-term investment strategy where an investor holds onto their digital assets (such as Bitcoin) through severe market volatility and price drops instead of panic-selling or active day trading.

CALCULATE: Why 21 Million

CALCULATE Add up every subsidy that will ever exist.

```
210,000 × (50 + 25 + 12.5 + …)
= 210,000 × 100 = 21,000,000 BTC   (almost)

in satoshis, rounded down at each halving:
= 2,099,999,997,690,000 sat
= 20,999,999.9769 BTC
```

Why the series stops

The subsidy is a whole number of satoshis. After 33 halvings it rounds down to zero, and only fees remain.

Where the money supply is decided

Not by a bank, not by a vote: by a rule every node checks in every coinbase.

21 million is the sum of the halvings — minus a few rounded satoshis.

The bug is that **Satoshi Nakamoto never programmed the Genesis block's coinbase transaction into UTXO database set.**

The first 50 coins exist, and can never move.

My video

The Genesis Block's Coins

Address	`1A1zP1eP5QGefi2DMPTfTL5SLmv7DivfNa`
Payments to it	66,961
Received in all	57.66531918 BTC
Ever spent	0



Block 0's coinbase paid 50 BTC to this address. Those 50 coins can never be spent: the first program never added them to its list of unspent outputs. People still send small amounts to it, as a **tribute**. Nothing has ever left.

New Word: Difficulty Adjustment

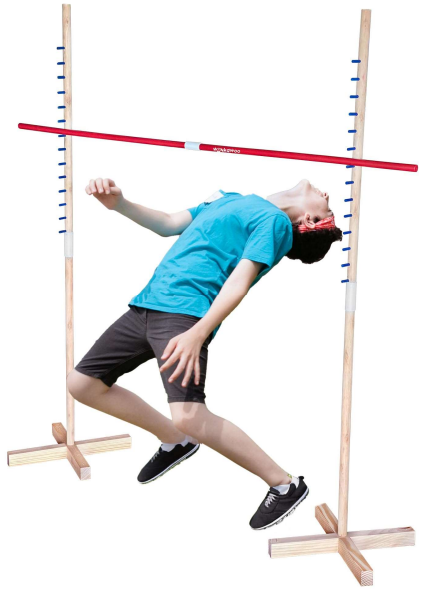
New word · Difficulty adjustment

Every 2,016 blocks — about two weeks — every node recomputes the target from how fast the last 2,016 blocks came. Too fast: the target goes down (harder). Too slow: up (easier).

That is why Bitcoin averages about 10 minutes per block (Lecture 3 measured 9.6), however many machines join.

My video

Every 2,016 blocks the target is retuned so that blocks keep coming every ten minutes.



Last adjustment	block 971,712, 16 Oct 2026: difficulty **+5.92 %**
This period so far	7.9 % done, average gap **579 s**
Next adjustment	block 973,728, about 30 Oct 2026; estimate **+2.0 %**

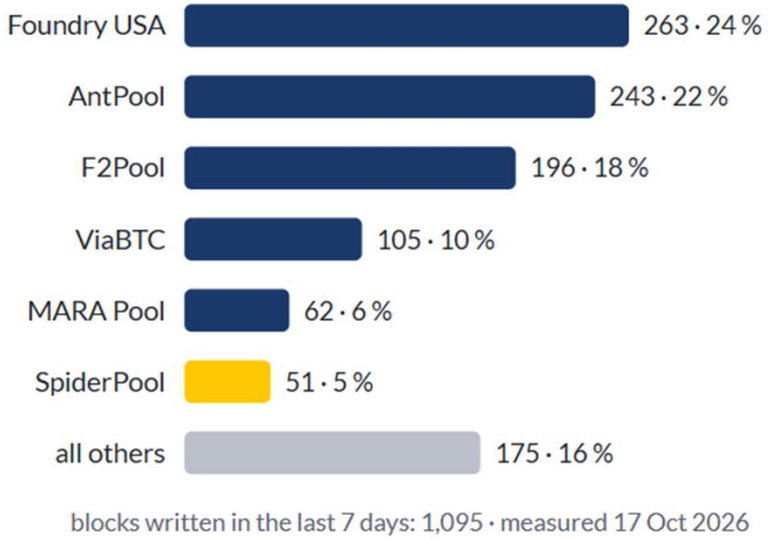
One Question Left

The danger	What if two miners win at once — or one gets too strong?

Who wins when the race is close?

One question is about people:
what if two miners win at the same time — or one becomes too strong?

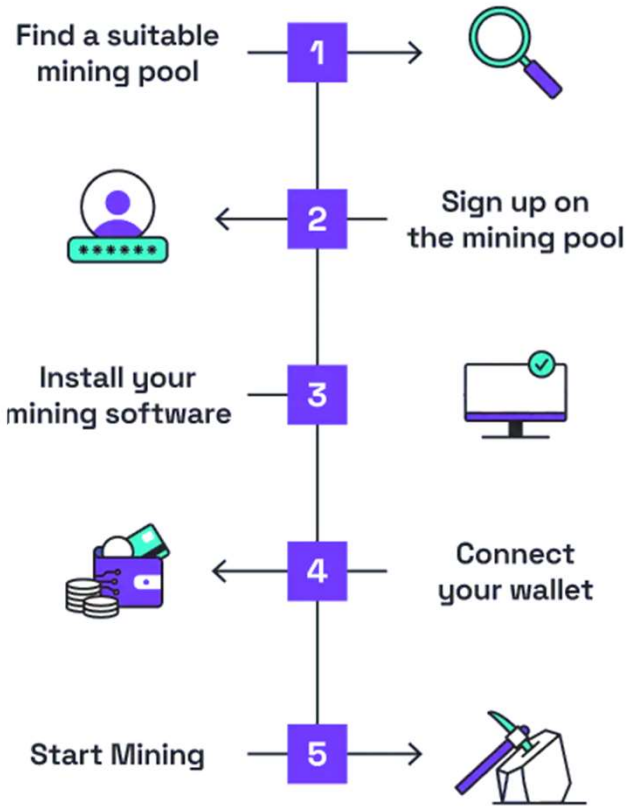
New Word: Mining Pool



Most blocks today come from a handful of pools.

📘 New word · Mining pool

A group of miners who search together and share each reward by the work each one did — a steady small income instead of a rare big win.



My video

Whitepaper vs Mainnet

The 2008 whitepaper: "Proof-of-work is essentially one-CPU-one-vote."
Last week: three pools wrote 64 % of all blocks.
Each pool is many independent miners, who can leave — next slides.

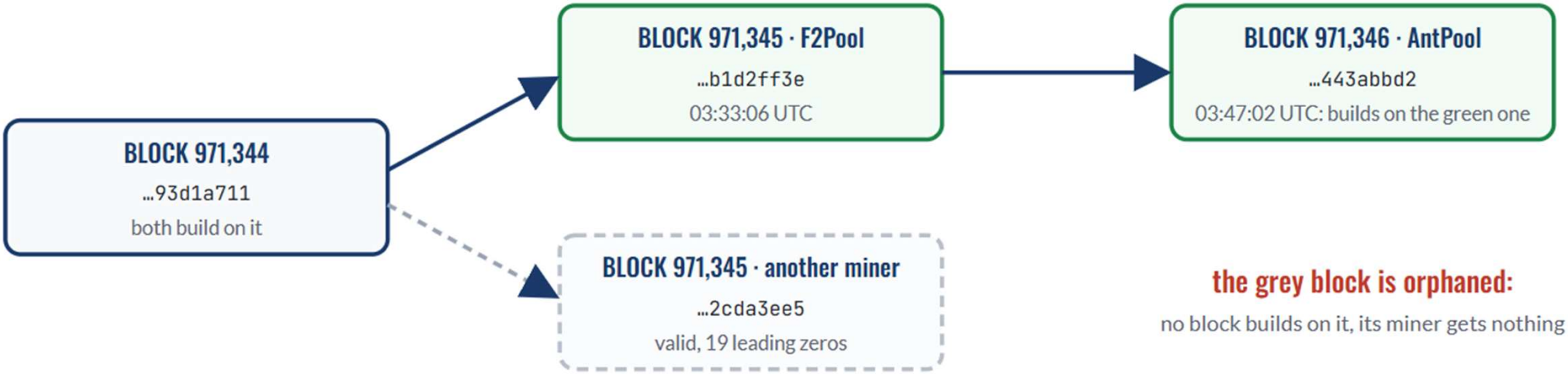
New Word: Orphan Block

New word · Orphan block

A valid block that loses the race: another block at the same height was built on instead. Its transactions go back to the mempool, unless the winner holds them too.

A real one: on 2026-10-14, two blocks 971,345 were found at about the same time. Some nodes heard of one first, some of the other — until, 14 minutes later, AntPool built on F2Pool's and every node followed.

My video



When two blocks tie, the next block decides; the loser is orphaned.

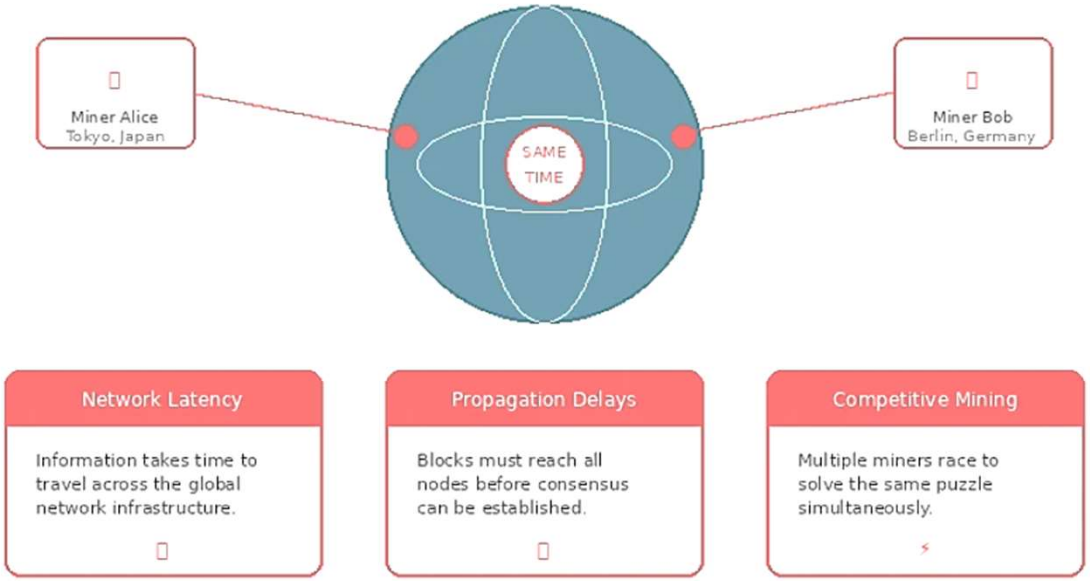
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My video



New Word: Confirmation

My video

New word · Confirmation

A payment has one confirmation when it is in a block, and one more for every block built on top. Each one makes it harder to undo.

Every new block on top is one more confirmation.

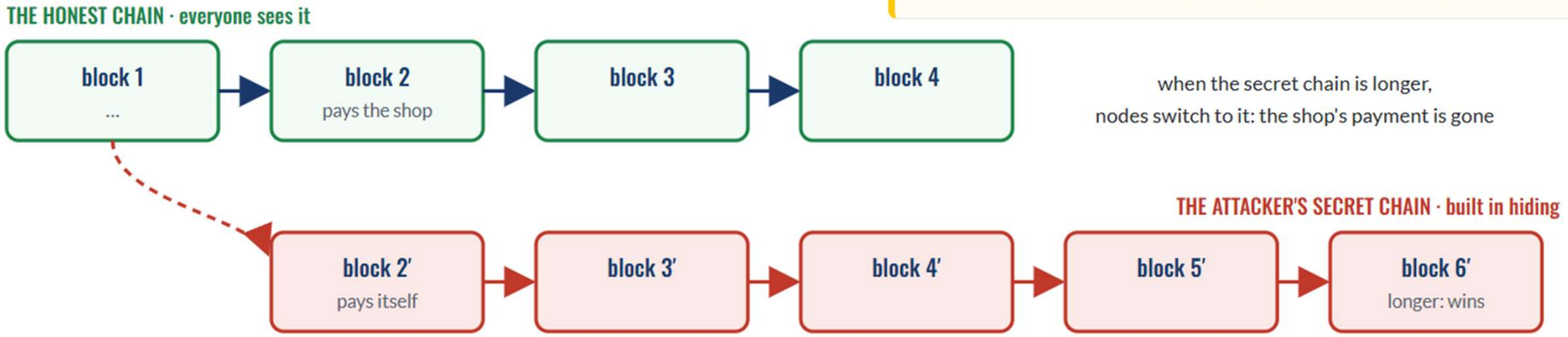
The rule every node follows: when there is a choice, take the chain with the most work in it. A payment deep under many blocks would need all of them redone.

CALCULATE Our payment from Lecture 6

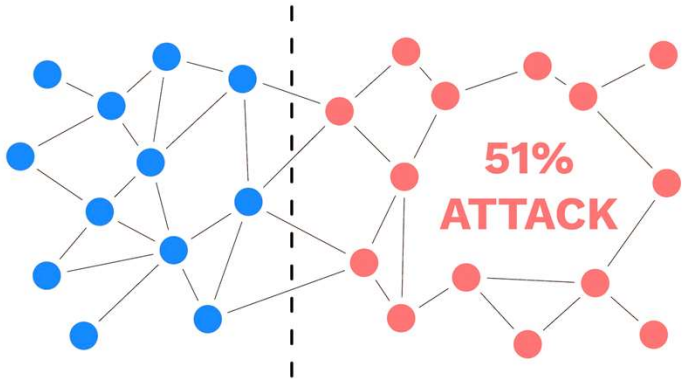
```
newest block      971,871
our block       - 968,391
                + 1
confirmations   = 3,481
```

New word · 51% attack

One miner, or a group, controls more than half of all hashing. It can then build a secret chain faster than everyone else, publish it, and replace recent blocks — undoing its own payments: double spending.



New Word: 51% Attack



What it still cannot do

Spend anybody else's coins: there is no signature (Lecture 4). Create coins beyond the rules: every node checks every coinbase. Rewrite old history cheaply: every block on top must be redone.

My video

📘 New word · 51% attack

One miner, or a group, controls more than half of all hashing. It can then build a secret chain faster than everyone else, publish it, and replace recent blocks — undoing its own payments: double spending.

THE HONEST CHAIN · everyone sees it

block 1 …

block 2 pays the shop

block 3

block 4

when the secret chain is longer, nodes switch to it: the shop's payment is gone

More than half of the work can rewrite recent blocks — never someone else's signature.

THE ATTACKER'S SECRET CHAIN · built in hiding

block 2′ pays itself

block 3′

block 4′

block 5′

block 6′ longer: wins

CALCULATE: Why Wait for Six Blocks?

CALCULATE A miner with share q of all hashing wins a given block with chance q. Six blocks in a row?

q	1 block	6 in a row: q^6
10 %	0.1	**0.000001** — about 1 in 1,000,000
30 %	0.3	**0.00073** — about 1 in 1,372
45 %	0.45	**0.0083** — about 1 in 120

So: wait

That is why many services wait for about six confirmations — one hour — before treating a large payment as final.

Each confirmation multiplies the attacker's odds by q.

Respect

Exchanges that trusted a few confirmations paid for the lesson; ordinary holders kept their coins.

My video

War Story: Bitcoin Gold, May 2018

Security is the cost of the work: little work, little security.

What	Bitcoin Gold — a separate, much smaller network that copied Bitcoin's design, with its own coin and far fewer miners
May 2018	someone rented enough machines to control more than half of its mining
Then	deposited coins at exchanges, traded them, withdrew — and replaced the blocks holding the deposits
Loss	over 18 million US dollars, reported, borne by exchanges



10/23/2026

SOC4190 2026 Yusupov